



NATIONAL INSTITUTE OF TECHNOLOGY, TIRUCHIRAPALLI
DEPARTMENT OF ENERGY & ENVIRONMENT
SUBJECT: ENERGY AND ENVIRONMENT (ENIR 11)

Date: 26/12/2025

Branch-Sec: ICE A & B, CSE-B
Time: 3 hours

Maximum Marks: 100

• All questions are compulsory. Only one answer booklet will be provided. Write your answers clearly, neatly, and precisely within the allotted space. Diagrams, wherever applicable, must be neat and properly labeled.

- 1) ☒ A) Discuss the present energy resources in India and explain the concept of energy sustainability. Compare conventional and non-conventional power generation with suitable examples. (6)
☒ B) Classify coal on the basis of its carbon content and mention the areas in India where each type can be found? (4)
☒ C) Explain the working principles, advantages, and disadvantages of any two conventional power plants used in India. (4)
- 2) ☒ A) What is the National Air Quality Index (AQI)? When was the National Air Quality Index India launched? Explain the various category based on AQI Index values and their associated health impacts? (10)
☒ B) Describe the sources and impacts of water pollution and explain the basic processes involved in the wastewater treatment plant? (10)
☒ C) Classify the sources of soil pollution and provide examples associated with each source? (5)
- 3) ☒ A) Explain the Betz limit? List out the parameters on which the wind power generation depends? Draw the arbitrary power curve of a 1 MW wind turbine compared to wind speed (kindly clearly mention the cut-in, cut-out and rated speed)? (10)
☒ B) In an area of 1000 m^2 where a large wind turbine is installed, the velocity of the wind is given as 10.3 m/s . The height of the turbine & diameter of its rotor is about 150 m & 130 m , respectively. Calculate the wind power density, the maximum power of the wind turbine and the maximum power output of a real wind turbine which has a power coefficient of 0.2 . (Assume air density is 1.456 kg/m^3) 795.5 6.22 2.11 (10)
☒ C) Determine the declination angle and hour angle at $9:30 \text{ AM}$ (solar time) on July 21, 1992. Assume the standard meridian for the location is 82.5°E and the place is ORION building NIT Trichy. The average temperature is 24°C and the relative humidity is 68% . -37.5 20.2 22.02 (10)
- 4) ☒ A) Write short notes on solar energy applications in day-to-day life and safety aspects while handling solar devices. (8)
☒ B) What are the different types of solar water heating system are there briefly write the various major thermal applications of solar energy? (6)
- 5) ☒ A) Draw a chart (no description required) showing various pathways available for biomass conversion into fuel/heat/power. (6)
☒ B) Explain the steps involved in the proximate analysis of biomass and describe the significance of each step in determining its properties. (6)
☒ C) Write explanatory notes on: (i) Ocean Thermal Energy Conversion (OTEC) (ii) Tidal Energy? (8)

The End
